

Prop. Let $p \in S$ be a point in the regular surface S .

For two Parametrization $X:U \rightarrow S$ and $Y:V \rightarrow S$ such that $p \in X[U] \cap Y[V] = W$, the change of coordinates $h = X^{-1} \circ Y: Y^{-1}[W] \rightarrow X^{-1}[W]$ is diffeomorphism.